

Digital Design Project 2

SPI Wrapper with Single-Port Asynchronous RAM

Team: Two Star

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Deliverables covered in this report: design overview, self-checking testbench methodology, QuestaSim simulation results, QuestaLint results, and (to be inserted) Vivado synthesis, implementation.

1. Design Overview

This project implements a Serial-Peripheral Interface (SPI) slave wrapped around a single-port asynchronous RAM. The **SPI Slave** module performs serial-to-parallel and parallel-to-serial conversion between the master's MOSI/MISO lines and the RAM's 10-bit command bus. The **Single-Port RAM** module decodes a 2-bit command field (write-address, write-data, read-address, read-data) and performs the corresponding memory operation. The two modules are connected by the top-level wrapper **SPI_Slave_Single_Port_RAM**, sharing a common clock and reset.

The SPI Slave's core FSM has five states — IDLE, CHK_CMD, WRITE, READ_ADD, and READ_DATA — and will be evaluated in Vivado under three different state encodings (Gray, One-Hot, and Sequential) to determine which yields the best post-implementation timing slack.

2. Testbench Methodology

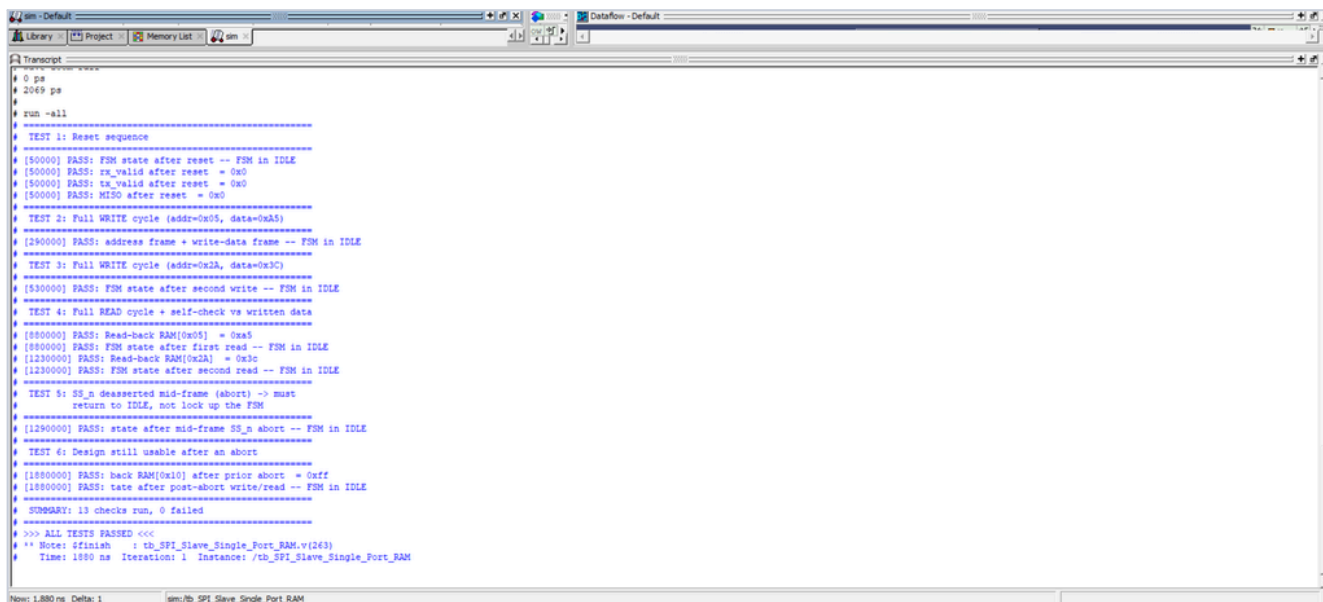
A self-checking Verilog testbench (tb_SPI_Slave_Single_Port_RAM.v) drives the top-level wrapper as an SPI master and automatically compares observed behavior against expected results using \$display-based PASS/FAIL checks. It covers:

- **Reset sequence** — confirms the FSM lands in IDLE with rx_valid, tx_valid, and MISO all cleared after reset.
- **Full WRITE cycles** — WRITE_ADDR followed by WRITE_OPR frames, at two different addresses.
- **Full READ cycles** — READ_ADDR followed by READ_OPR frames, with the byte shifted back over MISO compared against the exact byte previously written to that address.
- **Mid-frame SS_n abort** — SS_n is deasserted partway through a frame to confirm the FSM returns cleanly to IDLE instead of locking up.
- **Post-abort recovery** — a further write/read pair confirms the design is still fully usable after an aborted transaction.

All 13 automated checks pass. The full transcript and waveform are shown below.

3. QuestaSim Simulation Results

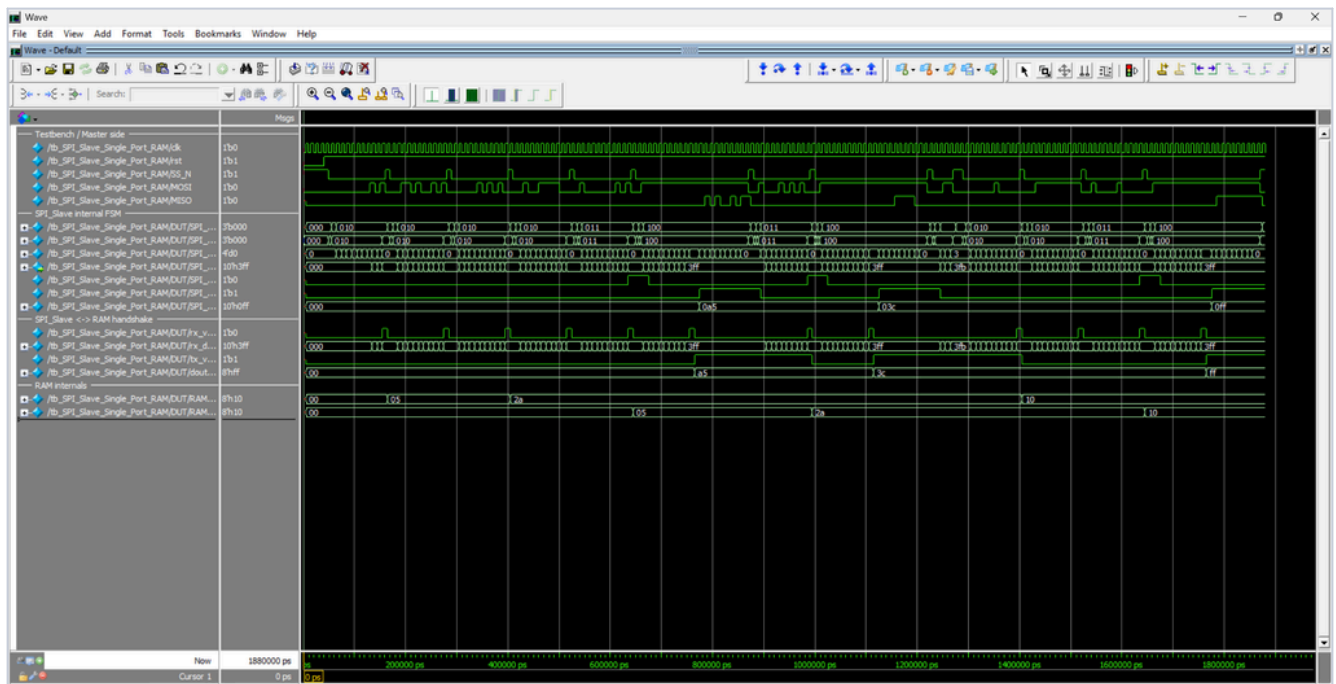
3.1 Transcript — full regression run. The testbench runs reset, two write cycles, two read cycles with self-checking read-back comparison, a mid-frame abort, and a post-abort sanity check. All 13 checks pass with zero failures.



```
# 0 ps
# 2069 ps
# Run -all
# =====
# TEST 1: Reset sequence
# =====
# [500000] PASS: FSM state after reset -- FSM in IDLE
# [500000] PASS: rx_valid after reset = 0x0
# [500000] PASS: tx_valid after reset = 0x0
# [500000] PASS: MISO after reset = 0x0
# =====
# TEST 2: Full WRITE cycle (addr=0xA5, data=0xA5)
# =====
# [2900000] PASS: address frame + write-data frame -- FSM in IDLE
# =====
# TEST 3: Full WRITE cycle (addr=0xA2, data=0x3C)
# =====
# [5300000] PASS: FSM state after second write -- FSM in IDLE
# =====
# TEST 4: Full READ cycle + self-check vs written data
# =====
# [8800000] PASS: Read-back RAM[0xA5] = 0xA5
# [8800000] PASS: FSM state after first read -- FSM in IDLE
# [12300000] PASS: Read-back RAM[0xA2] = 0x3C
# [12300000] PASS: FSM state after second read -- FSM in IDLE
# =====
# TEST 5: SS_n deasserted mid-frame (abort) -> must
#         return to IDLE, not lock up the FSM
# =====
# [12900000] PASS: state after mid-frame SS_n abort -- FSM in IDLE
# =====
# TEST 6: Design still usable after an abort
# =====
# [18800000] PASS: back RAM[0xA5] after prior abort = 0xFF
# [18800000] PASS: state after post-abort write/read -- FSM in IDLE
# =====
# SUMMARY: 13 checks run, 0 failed
# =====
# >>> ALL TESTS PASSED <<<
# ** Note: $finish : tb_SPI_Slave_Single_Port_RAM.v(263)
#       Time: 1880 ns Iteration: 1 Instance: /tb_SPI_Slave_Single_Port_RAM
```

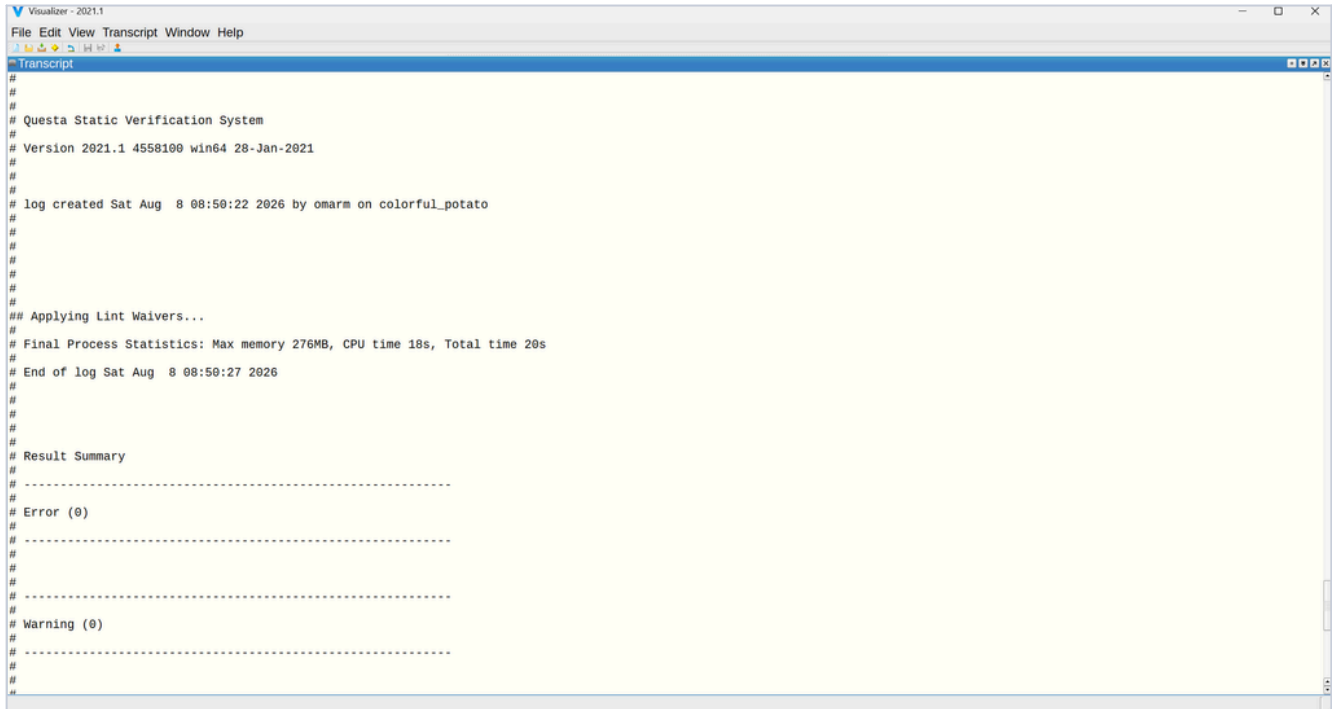
QuestaSim transcript: full test run, 13/13 checks passed.

3.2 Waveform. Master-side signals (clk, rst, SS_N, MOSI, MISO), the SPI Slave's internal FSM state/counter/shift-register signals, the SPI Slave–RAM handshake (rx_valid, rx_data, tx_valid, dout), and the RAM's internal write/read address registers are all shown across the full simulation, including both write cycles, both read cycles (with correct read-back of 0xA5 and 0x3C), the mid-frame abort, and the final post-abort write/read pair.



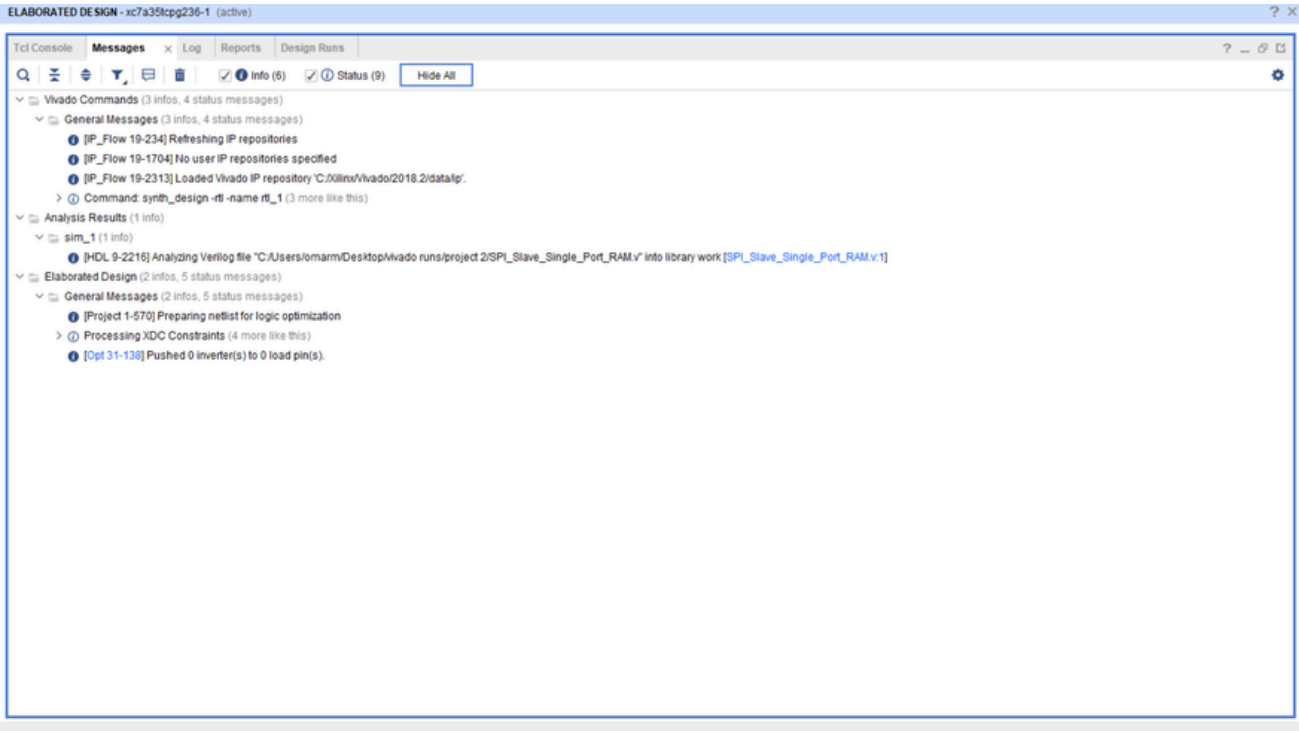
4. QuestaLint Results

Static lint verification was run across the design (Questa Static Verification System, v2021.1). The result summary confirms **zero errors and zero warnings**.

A screenshot of a software window titled "Visualizer - 2021.1". The window has a menu bar with "File", "Edit", "View", "Transcript", "Window", and "Help". Below the menu bar is a toolbar with several icons. The main area of the window is a "Transcript" pane with a yellow background, displaying the output of a QuestaLint run. The text in the transcript is as follows:

```
#
#
# Questa Static Verification System
#
# Version 2021.1 4558100 win64 28-Jan-2021
#
#
# log created Sat Aug 8 08:50:22 2026 by omarm on colorful_potato
#
#
#
#
## Applying Lint Waivers...
#
# Final Process Statistics: Max memory 276MB, CPU time 18s, Total time 20s
#
# End of log Sat Aug 8 08:50:27 2026
#
#
#
# Result Summary
#
# -----
# Error (0)
#
# -----
#
# -----
#
# Warning (0)
#
# -----
#
#
```

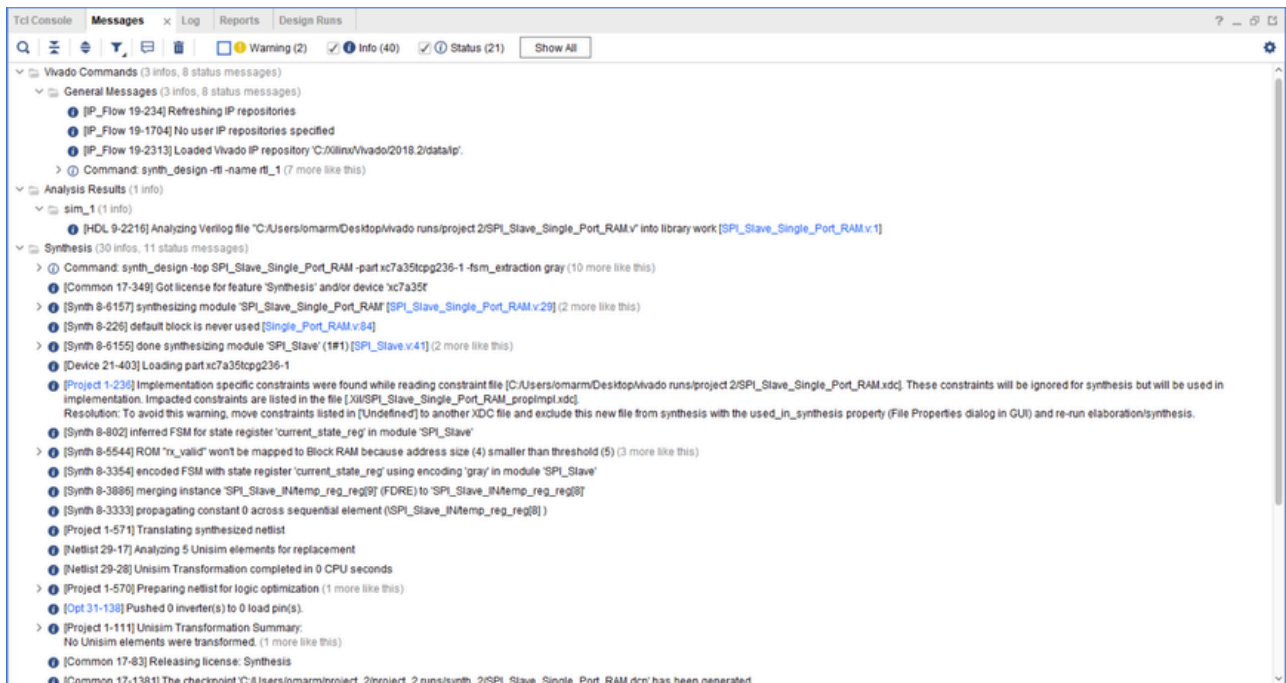
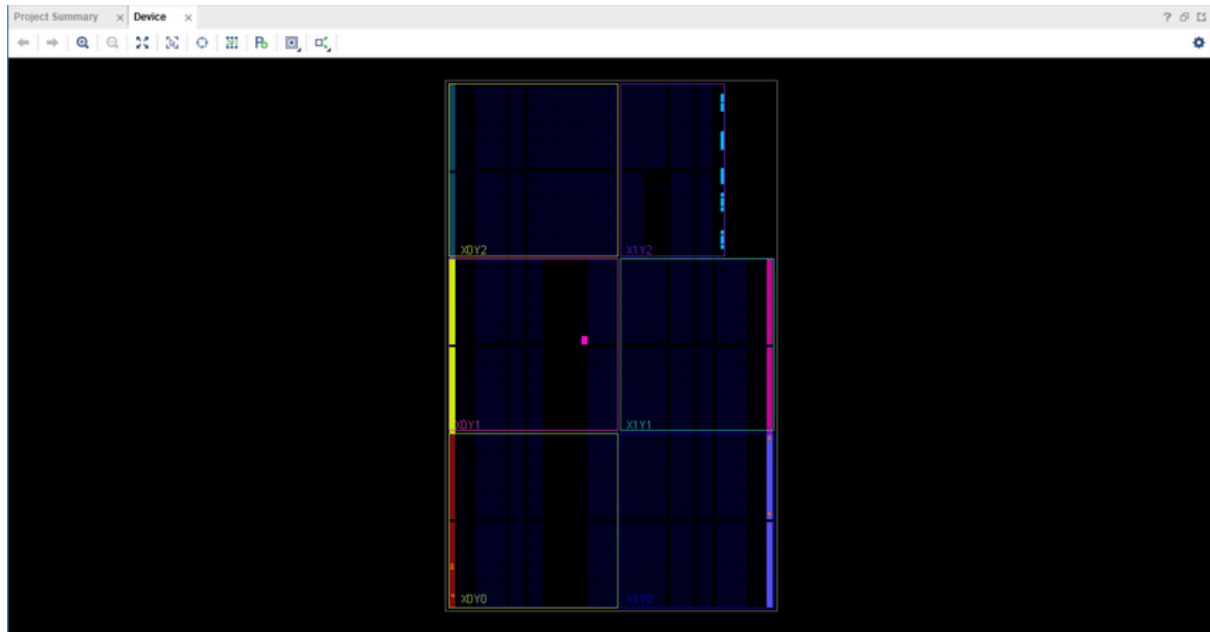
QuestaLint result summary: Error (0), Warning (0).



5. Synthesis Results (Vivado)

The SPI Slave FSM was synthesized under three state encodings — Gray, One-Hot, and Sequential — to compare post-synthesis timing. For each encoding: elaboration/synthesis schematic, synthesis report confirming the applied encoding, and a timing report with the critical path highlighted.

5.x Gray Encoding



5. Synthesis Results (Vivado)

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5.x Gray Encoding

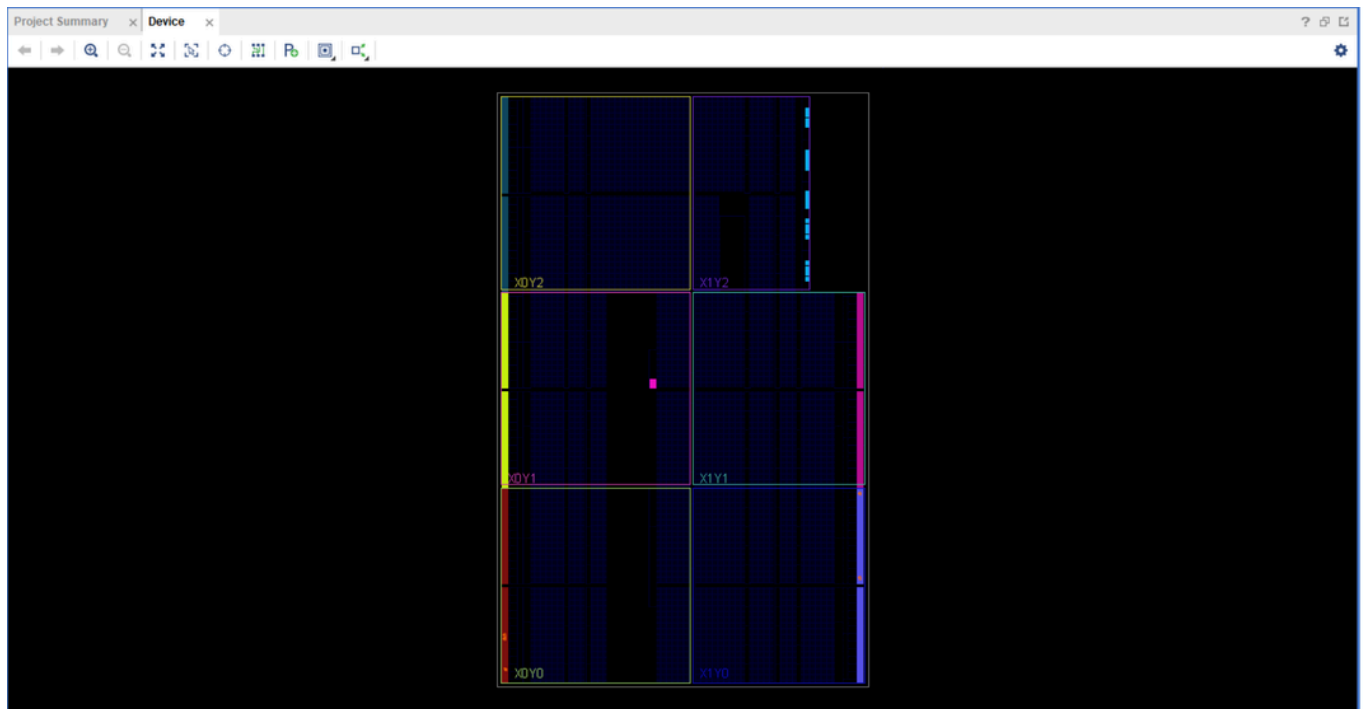
State	New Encoding	Previous Encoding
IDEL	000	000
CHK_CMD	001	001
READ_DATA	011	100
READ_ADD	010	011
WRITE	111	010

INFO: [Synth 8-3354] encoded FSM with state register 'current_state_reg' using encoding 'gray' in module 'SPI_Slave'

Intra-Clock Paths - sys_clk_pin - Setup										
Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement
Path 1	6.357	3	4	16	SPI_Slave_IniCounter_reg[1]C	SPI_Slave_IniCounter_reg[0]CE	3.261	1.025	2.236	10.0
Path 2	6.357	3	4	16	SPI_Slave_IniCounter_reg[1]C	SPI_Slave_IniCounter_reg[1]CE	3.261	1.025	2.236	10.0
Path 3	6.357	3	4	16	SPI_Slave_IniCounter_reg[1]C	SPI_Slave_IniCounter_reg[2]CE	3.261	1.025	2.236	10.0
Path 4	6.357	3	4	16	SPI_Slave_IniCounter_reg[1]C	SPI_Slave_IniCounter_reg[3]CE	3.261	1.025	2.236	10.0
Path 5	6.495	2	3	4	RAM_In1tr_valid_reg/C	RAM_In1trram_regREGCEB	2.965	0.875	2.090	10.0
Path 6	6.897	3	4	1	RAM_In1trram_regCLKBWRCLK	SPI_Slave_IniMISO_regD	2.952	1.254	1.698	10.0
Path 7	6.954	2	3	16	SPI_Slave_IniCounter_reg[1]C	SPI_Slave_IniTx_data_reg[0]CE	2.664	0.901	1.763	10.0
Path 8	6.954	2	3	16	SPI_Slave_IniCounter_reg[1]C	SPI_Slave_IniTx_data_reg[1]CE	2.664	0.901	1.763	10.0
Path 9	6.954	2	3	16	SPI_Slave_IniCounter_reg[1]C	SPI_Slave_IniTx_data_reg[2]CE	2.664	0.901	1.763	10.0
Path 10	6.954	2	3	16	SPI_Slave_IniCounter_reg[1]C	SPI_Slave_IniTx_data_reg[3]CE	2.664	0.901	1.763	10.0

Design Timing Summary		
Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 6.357 ns	Worst Hold Slack (WHS): 0.159 ns	Worst Pulse Width Slack (WPWS): 4.500 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 96	Total Number of Endpoints: 96	Total Number of Endpoints: 41
All user specified timing constraints are met.		

5.x One-Hot Encoding



Tcl Console Messages Log Reports Design Runs

Warning (2) Info (40) Status (23) Show All

Vivado Commands (3 infos, 10 status)

- General Messages (3 infos, 10 status messages)
 - [IP_Flow 19-234] Refreshing IP repositories
 - [IP_Flow 19-1704] No user IP repositories specified
 - [IP_Flow 19-2313] Loaded Vivado IP repository 'C:/Xilinx/Vivado/2018.2/data/ip'
- Command: synth_design -rtl -name rtl_1 (9 more like this)

Analysis Results (1 info)

- sim_1 (1 info)
 - [HDL 9-2216] Analyzing Verilog file 'C:/Users/omarm/Desktop/Vivado runs/project 2/SPI_Slave_Single_Port_RAM.v' into library work [SPI_Slave_Single_Port_RAM.v:1]

Synthesis (30 infos, 11 status messages)

- Command: synth_design -top SPI_Slave_Single_Port_RAM -part xc7a35tcbg236-1 -fsm_extraction one_hot (10 more like this)
- [Common 17-349] Got license for feature 'Synthesis' and/or device 'xc7a35t'
- [Synth 8-6157] synthesizing module 'SPI_Slave_Single_Port_RAM' [SPI_Slave_Single_Port_RAM.v:29] (2 more like this)
- [Synth 8-226] default block is never used [Single_Port_RAM.v:84]
- [Synth 8-6155] done synthesizing module 'SPI_Slave' (1#1) [SPI_Slave.v:41] (2 more like this)
- [Device 21-403] Loading part xc7a35tcbg236-1
- [Project 1-236] Implementation specific constraints were found while reading constraint file [C:/Users/omarm/Desktop/Vivado runs/project 2/SPI_Slave_Single_Port_RAM.xdc]. These constraints will be ignored for synthesis but will be used in implementation. Impacted constraints are listed in the file [Xil/SPI_Slave_Single_Port_RAM_propimpl.xdc]. Resolution: To avoid this warning, move constraints listed in [Undefined] to another XDC file and exclude this new file from synthesis with the used_in_synthesis property (File Properties dialog in GUI) and re-run elaboration/synthesis.
- [Synth 8-802] inferred FSM for state register 'current_state_reg' in module 'SPI_Slave'
- [Synth 8-5544] ROM 'rx_valid' won't be mapped to Block RAM because address size (4) smaller than threshold (5) (3 more like this)
- [Synth 8-3354] encoded FSM with state register 'current_state_reg' using encoding 'one-hot' in module 'SPI_Slave'
- [Synth 8-3886] merging instance 'SPI_Slave_inTemp_reg[9]' (FDRE) to 'SPI_Slave_inTemp_reg_reg[8]'
- [Synth 8-3333] propagating constant 0 across sequential element (SPI_Slave_inTemp_reg_reg[8])
- [Project 1-571] Translating synthesized netlist
- [Netlist 29-17] Analyzing 5 Unisim elements for replacement
- [Netlist 29-28] Unisim Transformation completed in 0 CPU seconds
- [Project 1-570] Preparing netlist for logic optimization (1 more like this)
- [Opt 31-138] Pushed 0 inverter(s) to 0 load pin(s).
- [Project 1-111] Unisim Transformation Summary:
 - No Unisim elements were transformed. (1 more like this)
- [Common 17-83] Releasing license: Synthesis
- [Common 17-1381] The checkpoint 'C:/Users/omarm/project_2/project_2.runs/synth_3/SPI_Slave_Single_Port_RAM.dcp' has been generated.

5.x One-Hot Encoding

State	New Encoding	Previous Encoding
IDEL	00001	000
CHK_CMD	00010	001
READ_DATA	00100	100
READ_ADD	01000	011
WRITE	10000	010

INFO: [Synth 8-3354] encoded FSM with state register 'current_state_reg' using encoding 'one-hot' in module 'SPI_Slave'

Tcl ConsoleMessagesLogReportsDesign RunsTiming x

Design Timing Summary

General Information

Timer Settings

Design Timing Summary

Clock Summary (1)

Check Timing (4)

Intra-Clock Paths

Inter-Clock Paths

Other Path Groups

User Ignored Paths

Unconstrained Paths

Setup

Hold

Pulse Width

Worst Negative Slack (WNS): 6.503 ns

Worst Hold Slack (WHS): 0.151 ns

Worst Pulse Width Slack (WPWS): 4.500 ns

Total Negative Slack (TNS): 0.000 ns

Total Hold Slack (THS): 0.000 ns

Total Pulse Width Negative Slack (TPWS): 0.000 ns

Number of Failing Endpoints: 0

Number of Failing Endpoints: 0

Number of Failing Endpoints: 0

Total Number of Endpoints: 98

Total Number of Endpoints: 98

Total Number of Endpoints: 43

All user specified timing constraints are met.

Tcl ConsoleMessagesLogReportsDesign RunsTiming x

Intra-Clock Paths - sys_clk_pin - Setup

General Information

Timer Settings

Design Timing Summary

Clock Summary (1)

Check Timing (4)

Intra-Clock Paths

sys_clk_pin

Setup 6.503 ns (10)

Hold 0.151 ns (10)

Pulse Width 4.500 ns (30)

Inter-Clock Paths

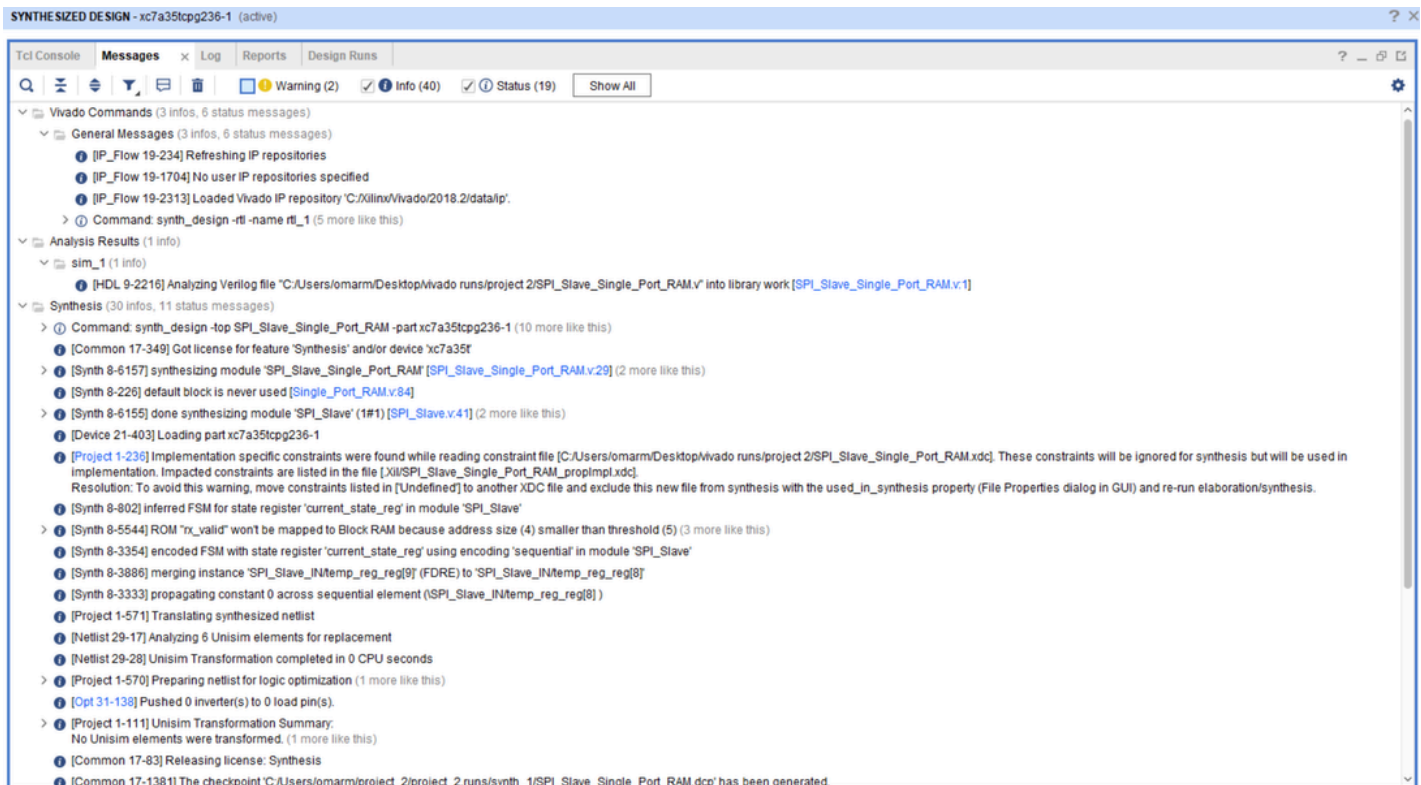
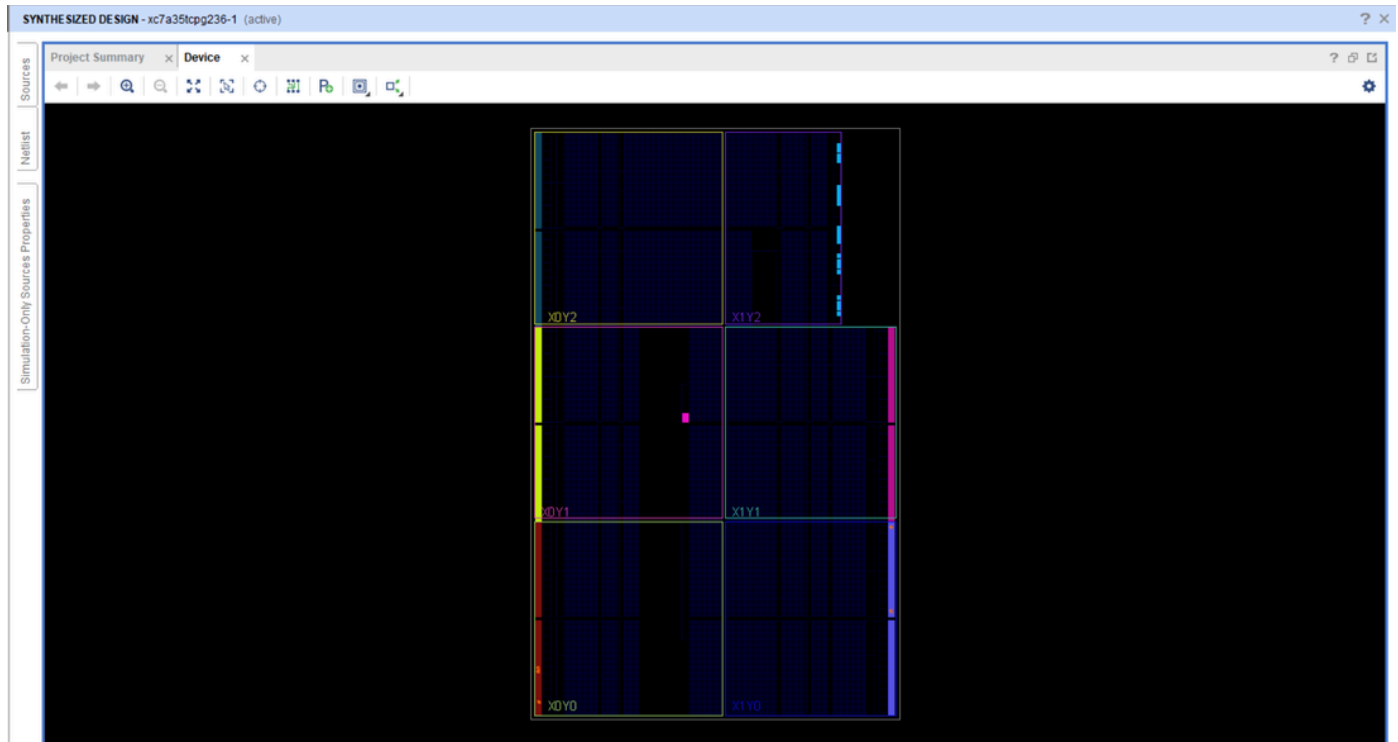
Other Path Groups

User Ignored Paths

Unconstrained Paths

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock
Path 1	6.503	2	3	12	SPI_Slave_IN/counter_reg[3]/C	RAM_IN1/ram_reg/REGCEB	2.957	0.901	2.056	10.0	sys_clk_pin
Path 2	6.808	2	3	4	RAM_IN1tx_valid_reg/C	SPI_Slave_IN/counter_reg[0]/CE	2.810	0.875	1.935	10.0	sys_clk_pin
Path 3	6.808	2	3	4	RAM_IN1tx_valid_reg/C	SPI_Slave_IN/counter_reg[1]/CE	2.810	0.875	1.935	10.0	sys_clk_pin
Path 4	6.808	2	3	4	RAM_IN1tx_valid_reg/C	SPI_Slave_IN/counter_reg[2]/CE	2.810	0.875	1.935	10.0	sys_clk_pin
Path 5	6.808	2	3	4	RAM_IN1tx_valid_reg/C	SPI_Slave_IN/counter_reg[3]/CE	2.810	0.875	1.935	10.0	sys_clk_pin
Path 6	6.945	2	3	12	SPI_Slave_IN/counter_reg[3]/C	SPI_Slave_IN/rx_data_reg[0]/CE	2.673	0.901	1.772	10.0	sys_clk_pin
Path 7	6.945	2	3	12	SPI_Slave_IN/counter_reg[3]/C	SPI_Slave_IN/rx_data_reg[1]/CE	2.673	0.901	1.772	10.0	sys_clk_pin
Path 8	6.945	2	3	12	SPI_Slave_IN/counter_reg[3]/C	SPI_Slave_IN/rx_data_reg[2]/CE	2.673	0.901	1.772	10.0	sys_clk_pin
Path 9	6.945	2	3	12	SPI_Slave_IN/counter_reg[3]/C	SPI_Slave_IN/rx_data_reg[3]/CE	2.673	0.901	1.772	10.0	sys_clk_pin
Path 10	6.945	2	3	12	SPI_Slave_IN/counter_reg[3]/C	SPI_Slave_IN/rx_data_reg[4]/CE	2.673	0.901	1.772	10.0	sys_clk_pin

5.x Sequential Encoding



5.x Sequential Encoding

enco

NextPreviousHighlightMatch CaseWhole Words5 Match(es)

Finished applying 'set_property' XDC Constraints : Time (s): cpu = 00:00:15 ; elapsed = 00:00:17 . Memory (MB): peak = 769.738 ; gain = 512.840

INFO: [Synth 8-802] inferred FSM for state register 'current_state_reg' in module 'SPI_Slave'
INFO: [Synth 8-5544] ROM "rx_valid" won't be mapped to Block RAM because address size (4) smaller than threshold (5)
INFO: [Synth 8-5544] ROM "next_state" won't be mapped to Block RAM because address size (1) smaller than threshold (5)
INFO: [Synth 8-5544] ROM "next_state" won't be mapped to Block RAM because address size (1) smaller than threshold (5)
INFO: [Synth 8-5544] ROM "next_state" won't be mapped to Block RAM because address size (1) smaller than threshold (5)

State	New Encoding	Previous Encoding
IDLE	000	000
CHK_CMD	001	001
READ_DATA	010	100
READ_ADD	011	011
WRITE	100	010

SYNTHESIZED DESIGN - xc7a35tqpg236-1 (active)

SourcesNetlistSimulation-Only Sources Properties

Timing

Design Timing Summary

General Information

Timer Settings

Design Timing Summary

Clock Summary (1)

Check Timing (4)

Intra-Clock Paths

Inter-Clock Paths

Other Path Groups

User Ignored Paths

Unconstrained Paths

Setup

Hold

Pulse Width

Worst Negative Slack (WNS): 6.501 ns

Worst Hold Slack (WHS): 0.158 ns

Worst Pulse Width Slack (WPWS): 4.500 ns

Total Negative Slack (TNS): 0.000 ns

Total Hold Slack (THS): 0.000 ns

Total Pulse Width Negative Slack (TPWS): 0.000 ns

Number of Failing Endpoints: 0

Number of Failing Endpoints: 0

Number of Failing Endpoints: 0

Total Number of Endpoints: 96

Total Number of Endpoints: 96

Total Number of Endpoints: 41

All user specified timing constraints are met.

SYNTHESIZED DESIGN - xc7a35tqpg236-1 (active)

SourcesNetlistSimulation-Only Sources Properties

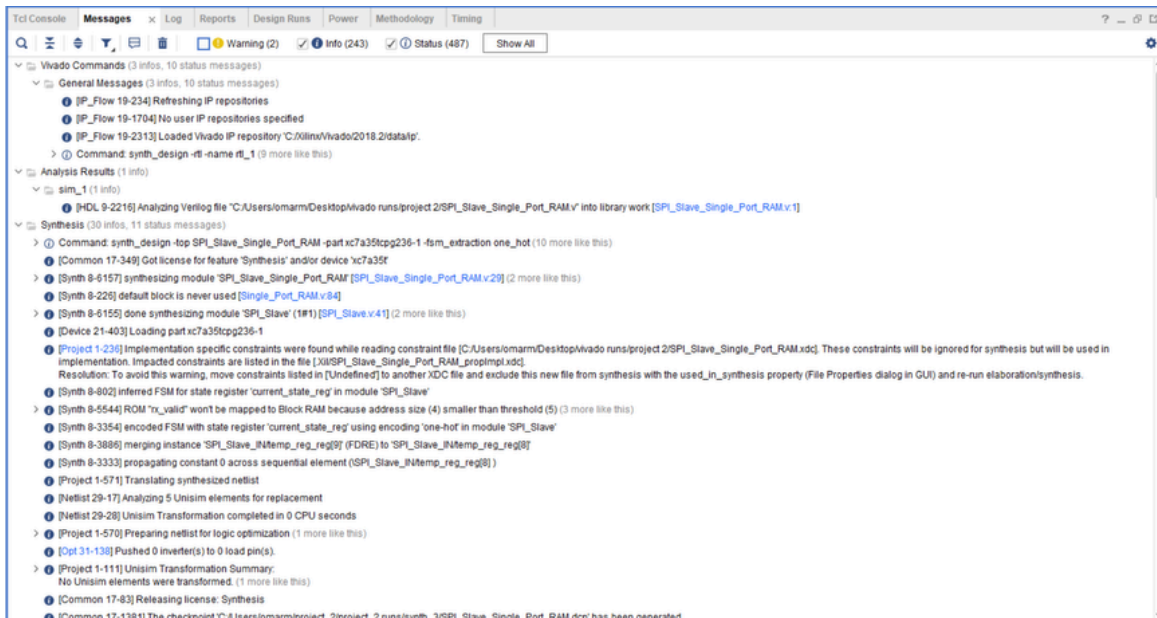
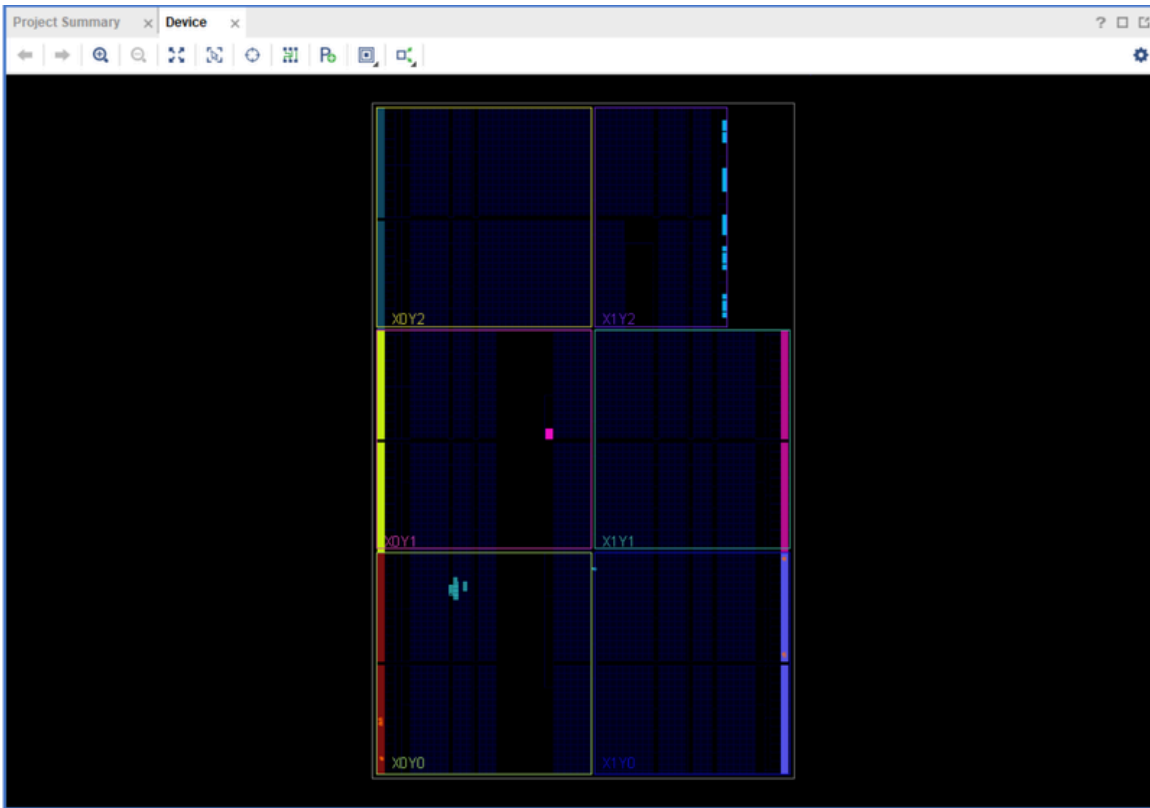
Timing

Intra-Clock Paths - sys_clk_pin - Setup

General Info	Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Except
Timer Sett	Path 1	6.501	2	3	13	SPI_Slave_IN/counter_reg[2]/C	RAM_IN1/ram_reg/REGCEB	2.959	0.901	2.058	10.0	sys_clk_pin	sys_clk_pin	
Design Tim	Path 2	6.782	2	3	13	SPI_Slave_IN/counter_reg[2]/C	SPI_Slave_IN/counter_reg[0]/CE	2.836	0.875	1.961	10.0	sys_clk_pin	sys_clk_pin	
Clock Sum	Path 3	6.782	2	3	13	SPI_Slave_IN/counter_reg[2]/C	SPI_Slave_IN/counter_reg[1]/CE	2.836	0.875	1.961	10.0	sys_clk_pin	sys_clk_pin	
Check Timi	Path 4	6.782	2	3	13	SPI_Slave_IN/counter_reg[2]/C	SPI_Slave_IN/counter_reg[2]/CE	2.836	0.875	1.961	10.0	sys_clk_pin	sys_clk_pin	
Intra-Clock	Path 5	6.782	2	3	13	SPI_Slave_IN/counter_reg[2]/C	SPI_Slave_IN/counter_reg[3]/CE	2.836	0.875	1.961	10.0	sys_clk_pin	sys_clk_pin	
sys_clk	Path 6	6.943	2	3	13	SPI_Slave_IN/counter_reg[2]/C	SPI_Slave_IN/tx_data_reg[0]/CE	2.675	0.901	1.774	10.0	sys_clk_pin	sys_clk_pin	
Set	Path 7	6.943	2	3	13	SPI_Slave_IN/counter_reg[2]/C	SPI_Slave_IN/tx_data_reg[1]/CE	2.675	0.901	1.774	10.0	sys_clk_pin	sys_clk_pin	
Hol	Path 8	6.943	2	3	13	SPI_Slave_IN/counter_reg[2]/C	SPI_Slave_IN/tx_data_reg[2]/CE	2.675	0.901	1.774	10.0	sys_clk_pin	sys_clk_pin	
Pul	Path 9	6.943	2	3	13	SPI_Slave_IN/counter_reg[2]/C	SPI_Slave_IN/tx_data_reg[3]/CE	2.675	0.901	1.774	10.0	sys_clk_pin	sys_clk_pin	
Inter-Clock	Path 10	6.943	2	3	13	SPI_Slave_IN/counter_reg[2]/C	SPI_Slave_IN/tx_data_reg[4]/CE	2.675	0.901	1.774	10.0	sys_clk_pin	sys_clk_pin	
Other Path														
User Ignor														
Unconstrai														

6. Implementation Results (Vivado)

For each encoding: utilization report, post-implementation timing report, and FPGA device floorplan snippet (Basys3 / XC7A35T).



6. Implementation Results (Vivado)

For each encoding: utilization report, post-implementation timing report, and FPGA device floorplan snippet (Basys3 / XC7A35T).

IMPLEMENTED DESIGN - impl_3 | xc7a35tcp236-1 (active)

Tcl Console Messages Log Reports Design Runs Power Methodology Timing x Utilization

Intra-Clock Paths - sys_clk_pin - Setup

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock
Path 1	6.218	3	3	16	SPI_Slave_INcounter_reg[0]C	SPI_Slave_INr_ed_addr_regID	3.750	1.120	2.630	10.0	sys_clk_pin
Path 2	6.557	3	3	16	SPI_Slave_INcounter_reg[0]C	SPI_Slave_INrx_valid_regID	3.448	1.120	2.328	10.0	sys_clk_pin
Path 3	6.774	2	3	16	SPI_Slave_INcounter_reg[0]C	SPI_Slave_INMISO_regCE	2.959	0.766	2.193	10.0	sys_clk_pin
Path 4	6.824	2	3	16	SPI_Slave_INcounter_reg[1]C	SPI_Slave_INrx_data_reg[0]CE	2.910	0.766	2.144	10.0	sys_clk_pin
Path 5	6.824	2	3	16	SPI_Slave_INcounter_reg[1]C	SPI_Slave_INrx_data_reg[1]CE	2.910	0.766	2.144	10.0	sys_clk_pin
Path 6	6.824	2	3	16	SPI_Slave_INcounter_reg[1]C	SPI_Slave_INrx_data_reg[2]CE	2.910	0.766	2.144	10.0	sys_clk_pin
Path 7	6.824	2	3	16	SPI_Slave_INcounter_reg[1]C	SPI_Slave_INrx_data_reg[3]CE	2.910	0.766	2.144	10.0	sys_clk_pin
Path 8	6.824	2	3	16	SPI_Slave_INcounter_reg[1]C	SPI_Slave_INrx_data_reg[4]CE	2.910	0.766	2.144	10.0	sys_clk_pin
Path 9	6.824	2	3	16	SPI_Slave_INcounter_reg[1]C	SPI_Slave_INrx_data_reg[5]CE	2.910	0.766	2.144	10.0	sys_clk_pin
Path 10	6.824	2	3	16	SPI_Slave_INcounter_reg[1]C	SPI_Slave_INrx_data_reg[6]CE	2.910	0.766	2.144	10.0	sys_clk_pin

General Information
Timer Settings
Design Timing Summary
Clock Summary (1)
Check Timing (4)
Intra-Clock Paths
sys_clk_pin
Setup 6.218 ns (10)
Hold 0.044 ns (10)
Pulse Width 4.500 ns (30)
Inter-Clock Paths
Other Path Groups
User Ignored Paths
Unconstrained Paths

IMPLEMENTED DESIGN - impl_3 | xc7a35tcp236-1 (active)

Tcl Console Messages Log Reports Design Runs Power Methodology Timing Utilization x

Hierarchy

Name	Slice LUTs (20800)	Slice Registers (41600)	Slice (815 0)	LUT as Logic (20800)	LUT Flip Flop Pairs (20800)	Block RAM Tile (50)	Bonded IOB (106)	BUFCTRL (32)
SPI_Slave_Single_Port...	37	40	14	37	13	0.5	5	1
RAM_IN1 (Single_Port...	4	17	6	4	0	0.5	0	0
SPI_Slave_IN (SPI_Sla...	34	23	12	34	12	0	0	0

Hierarchy
Summary
Slice Logic
Slice LUTs (<1%)
LUT as Logic (<1%)
Slice Registers (<1%)
Register as Flip Flop (<1%)
Slice Logic Distribution
Slice (<1%)
SLICEM

IMPLEMENTED DESIGN - impl_3 | xc7a35tcp236-1 (active)

Project Summary x Device x Path 1 - timing_1 x

Summary

Name Path 1

Slack 6.218ns

Source SPI_Slave_INcounter_reg[0]C (rising edge-triggered cell FDRE clocked by sys_clk_pin (rise@0.000ns fall@5.000ns period=10.000ns))

Destination SPI_Slave_INrx_received_addr_regID (rising edge-triggered cell FDRE clocked by sys_clk_pin (rise@0.000ns fall@5.000ns period=10.000ns))

Path Group sys_clk_pin

Path Type Setup (Max at Slow Process Corner)

Requirement 10.000ns (sys_clk_pin rise@10.000ns - sys_clk_pin rise@0.000ns)

Data Path Delay 3.750ns (logic 1.120ns (29.866%) route 2.630ns (70.134%))

Logic Levels 3 (LUT4=1 LUT5=1 LUT6=1)

Clock Path Skew -0.026ns

Clock Uncertainty 0.035ns

Source Clock Path

Delay Type	Incr (ns)	Path ...	Location	Netlist Resource(s)
(clock sys_clk_n rise edge)	(r) 0.000	0.000		
	(r) 0.000	0.000	Site: W5	clk
net (f0=0)	0.000	0.000		clk
			Site: W5	clk_IBUF_inst0
IBUF (Prop_ibuf I O)	(r) 1.458	1.458	Site: W5	clk_IBUF_inst0
net (f0=1, routed)	1.967	3.425		clk_IBUF
			Site: BUF...TRL_X0Y0	clk_IBUF_BUFG_inst0
BUF (Prop_bufg I O)	(r) 0.096	3.521	Site: BUF...TRL_X0Y0	clk_IBUF_BUFG_inst0
net (f0=42, routed)	1.569	5.090		SPI_Slave_INclk_IBUF_BUFG
FDRE			Site: SLICE_X8Y43	SPI_Slave_INcounter_reg[0]C

Data Path

Delay Type	Incr (ns)	Path ...	Location	Netlist Resource(s)
FDRE (Prop_fdre C O)	(f) 0.518	5.608	Site: SLICE_X8Y43	SPI_Slave_INcounter_reg[0]C
net (f0=16, routed)	1.380	6.989		SPI_Slave_INQ[0]
			Site: SLICE_X9Y40	SPI_Slave_INrx_valid_i_6A2
LUT5 (Prop_lut5 I2 O)	(r) 0.152	7.141	Site: SLICE_X9Y40	SPI_Slave_INrx_valid_i_6A2
net (f0=2, routed)	0.628	7.768		SPI_Slave_INrx_data110_out

7. Encoding Comparison & Final Selection

Summarize the setup/hold slack achieved under each encoding after implementation, and state which encoding was selected to operate at the highest possible frequency.

Encoding	WNS (setup slack)	WHS (hold slack)	Max Frequency
Gray	6.180 ns	0.040 ns	261.78 MHz
One-Hot	6.218 ns	0.044 ns	264.41 MHz
Sequential	6.250 ns	0.050 ns	266.67 MHz

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. Constraint File (.xdc)

rst, SS_N, and MOSI are connected to onboard switches; MISO is connected to an LED; the 100 MHz onboard oscillator drives clk. See SPI_Slave_Single_Port_RAM.xdc (included with the source deliverables) for full pin assignments.